

SAP HANA Interview Questions

1. Compare SAP HANA & Oracle

Criteria	SAP HANA	Oracle
Strengths	Customization, customer service & functionality	Ease of use, quick deployment and flexibility
Compatibility to work with other databases	Very Good	Average
Comparison on costs	More cost for internal resources and training	Much lesser on both fronts

2. What is SAP HANA?

SAP HANA is an in-memory computing engine (IMCE) used for real-time processing of huge volumes of data and building and deploying real-world applications. Adopting the row-based and column-based DB technology, SAP HANA is an advanced relational DB product serviced by SAP SE. With this high-performance analytic (HANA) system, the big data exists on the main memory and not on the hard disk. It replaces the onus of data maintenance separately on the legacy system and simplifies the tasks of administrators in this digital world.

3. Explain the column and Row store in HANA?

HANA supports both type of data store in database. Row store is used when you need to use Select statement and no aggregations are performed.

Column store is used to perform aggregations and HANA Modeling is supported only on Column based tables.

4. Which companies are providing SAP HANA hardware appliances?

There are total 11 vendors for SAP HANA hardware appliances. Most common are –

- Dell
- IBM
- HP
- Cisco
- Lenovo

5. What is the use of SAP HANA studio?

HANA studio is an eclipsed based tool and provides support for development and administration in HANA system. You can perform HANA Modeling on the top of tables in database, Data provisioning, HANA Administration and various other activities using HANA studio.

SAP HANA Studio client is available for Windows XP, Windows Vista, and Windows 7 for 32 bit and 64 bit operating system.

6. What do you understand by SAP HANA In-Memory Computing Engine IMCE?

In-Memory concept of SAP HANA means that all the data is stored in RAM memory. A conventional database transfer data from memory in 5 milliseconds however SAP HANA In-memory takes 5 nanoseconds to read data.

SAP HANA uses multicore CPU architecture and stores data in row and column based storage in HANA database.

7. What is the development language used by SAP HANA?

C++

8. Which operating system is supported by HANA system?

Only operating system that is supported by HANA is Suse Linux Enterprise Server SP1/SP2 (SLES SP1/2).

9. What is the functional difference between Row and Column based storage? Where do we use Row based storage and column based storage?

Consider below table- FCTSales

Country	Product	Units Sold
England	iphone 6	107
India	Samsung Note 6	250
US	Lenovo A110	110

Row Based Storage –

- England
- Iphone6
- 107
- India
- Samsung Note 6
- 250
- US
- Lenovo A110
- 110

Column Based Storage –

- England
- India
- US
- Iphone6
- Samsung Note6

- Lenovo A110
- 107
- 250
- 110

10. Where to use Row based storage?

Select * from FCTSales where Country='US'

11. List advantages of using SAP HANA database.

- With the HANA technology, you can create gen-next applications giving effective and efficient results in the digital economy.
- By using single data-in memory, SAP HANA supports smooth transaction process and fault-tolerant analytics
- Easy and simple operations using an open-source, unified platform in the cloud
- High-level Data Integration to access massive amounts of data
- Advanced tools for in-depth analysis of present, past and the future.

12. Where to use Column based storage?

Select SUM(Units_sold) from FCTSales where Product='Lenovo A110'

13. What are the different components in SAP HANA Architecture? What is the use of Index server?

- Index Server
 - Name Server
 - Statistical Server
 - Preprocessor Server
 - XS Engine
 - SAP Host Agent
 - LM structure
 - SAP Solution Manager Diagnostic Agent
- Index server contains engine to process data in HANA database. These data engines are responsible to handle all SQL/MDX statement in HANA system. Index server also contains Session and Transaction Manager which is responsible to manage all running and completed transactions.

14. What is the use of Persistence layer in SAP HANA system?

Persistence layer provides inbuilt mechanism for disaster recovery in HANA system. It ensures that database is restored to most recent state in case of a system failure.

Persistence layer also manages data, transaction and configuration logs and backup of these files. Backups of data and log files are performed at save points and is normally scheduled every 5-10 minutes.

15. What are the different license keys type in HANA system? What is their validity?

- Temporary License key
- Permanent License Key

Temporary License keys are automatically installed when you install the HANA database. These keys are valid only for 90 days and you should request permanent license keys from SAP market place before expiry of this 90 days period after installation.

Permanent License keys are valid till the predefined expiration date. License keys specify amount of memory licensed to target HANA installation.

16. List the merits and demerits of using row-based tables.

Merits:

- No data approach can be faster than row-based if you want to analyze, process and retrieve one record at one time.
- Row-based tables are useful when there is specific demand of accessing complete record.
- It is preferred when the table consists of less number of rows.
- This data storage and processing approach is easier and effective without any aggregations and fast searching.

Demerits:

- The data retrieval and processing operations involve the complete row, even though all the information is not useful.

Go through this in-depth [SAP HANA training](#) and master the powerful columnar database now!

17. What are the different types of permanent license keys in HANA system?

There are two types of permanent License keys for HANA system –

Unenforced – If unenforced license key is installed and consumption of HANA system exceeds the license amount of memory, operation of SAP HANA is not affected in this case.

Enforced – If Enforced license key is installed and consumption of HANA system exceeds the license amount of memory, HANA system gets locked. If this situation occurs, HANA system has to be restarted or a new license key should be requested and installed.

While activating SAP HANA Modeling view, you get an error message –Repository: Encountered an error in repository runtime extension; Deploy Attribute View: SQL: transaction rolled back by an internal error: insufficient privilege: Not authorized

Grant SELECT privileges on schemas of the used data foundation tables to user “_SYS_REPO”

GRANT SELECT ON SCHEMA “<SCHEMA_NAME>” TO _SYS_REPO WITH GRANT OPTION

18. In SAP HANA Studio, what is use of different folders when you add a HANA system to Studio?

Backup –

It is used to perform for backup and recovery in SAP HANA system. You can check backup configuration details, run manual backup, to check last successful back performed, etc. for data and log backup.

Catalog –

This contains RDBMS objects like schemas, tables, views, procedures, etc. You can open SQL editor and design database objects

Content –

This is used to maintain design time repository

You can create new packages and design Information views in HANA system. Various views can be created under content tab to meet business requirement and to perform analytical reports on the top of the Modeling views.

Provisioning –

This is used for Smart data access to connect to other databases like HADOOP, TERADATA and SYBASE

Security –

This is used to define users and to assign roles. You can define various privileges on different users using Security tab. You can assign Database and Package privileges to different users to control the data access.

19. What is the difference between Open Data Preview and Open Definition?

Open Data Preview –

This is used to see the data stored in an object- table or a modeling view. When you open data preview, you get three options –

- Raw Data
- Distinct Values
- Analysis

Open Definition –

This is used to see the structure of the table – column name, column data type, keys, etc.

20. If you want to see all active alerts, description and priority, where this can be checked?

Go to Administration → alerts

In Administration tab, you can check system overview, landscape, volumes, configuration, system information, etc.

21. What is SAP HANA cockpit?

SAP HANA cockpit is a SAP Fiori Launchpad site that allows you with a single point-of-access to a range of Web-based applications for the online administration of SAP HANA. You access the SAP HANA cockpit through a Web browser.

To open SAP HANA Cockpit → Right click on HANA system in Studio → configuration and monitoring → open SAP HANA cockpit

22. What is the difference between Catalog and Content tab?

Catalog –

This contains RDBMS objects like schemas, tables, views, procedures, etc. You can open SQL editor and design database objects.

Content –

This is used to maintain design time repository. You can create new packages and design Information views in HANA system. Various views can be created under content tab to meet business requirement and to perform analytical reports on the top of the Modeling views.

23. Which Information view in SAP HANA is used to implement star schema queries?

Analytic View

24. What is difference between Copy and derived from option while creating a new SAP HANA Attribute information view?

Copy option allows you to copy an existing Information view and to make changes to it.

Derived option allows to create a copy of an existing view and you can't make any changes to it.

25. What is the use of Data Foundation, Star Join and Semantics?

Data Foundation – to add column base tables.

Star Join – to add other type of views.

Semantic – to define user parameters and to define measures and dimensions.

26. What are the different user parameters that can be defined in Semantic layer?

- Measure and Attribute
- Hierarchies
- Parameters/Variables
- New Calculated Column

27. What is a new calculated column in a HANA Modeling view?

New calculated column is defined as a column added on the fly in Analysis tab when a view is activated. This column doesn't exist at database level or in Data Foundation or Star Join level.

28. To improve the performance of Information view by validating joins and Cardinality, which option can be used?

You can select Switch to Performance Analysis Mode. This allows you to do Performance Analysis by validating correct join type and Cardinality.

29. What are the benefits of using Calculation view with Star join?

It simplifies the design process as allows you to select multiple measures from multiple fact tables.

You can implement 3NF using Star Join.

30. What is the use of parameters and variables in Semantic layer?

Variables are used as an explicit SQL filter directive for view consumers to filter the view data, based on attribute column values specified in variable UI prompt of a BI Client. When you go to data preview, variables allow users to pass the value of attribute defined in the variable.

You can select a single value, range or range in selection type.

31. What are the default nodes available in a Calculation view?

- Join
- Union
- Project
- Aggregation
- Rank.

32. What is the default node, if you select a Dimension Calculation view?

Projection

33. How do you check if an Information view is saved and activated?

There is a diamond mark on view name if it is not active.

34. Can you pass Attribute views and analytic views in a Calculation view with Star join?

No. In a Calculation view with Star Join, you can only use Dim Calculation views.

35. What is SAP HANA Analytic Privileges?

Analytic Privileges are used to limit access on HANA Information views. You can assign different types of right to different users on different component of a View in Analytic Privileges.

36. If you want that your Area Managers can only see data from their region and you want to assign this restriction for specific duration, how this can be achieved?

Using Analytic Privileges, you can add Region attribute and values to Attribute Restrictions and Time duration is defined in Privilege validity.

Analytic Privilege can be added to user profile in User and Roles under Security tab.

37. What are the different methods of data replication in HANA?

- SAP LT Replication SLT
- SAP Data Services
- Direct Extract Connection DXC

38. What are the different modes in data provision in HANA Studio?

- Load
- Replicate
- Suspend
- Resume

39. What do you understand by trusted RFC connection? Which transaction is used to create and configure trusted RFC?

On your source SAP system A1 you want to setup a trusted RFC to target system B1. When it is done it would mean that when you are logged onto A1 and your user has enough Authorization in B1, you can use the RFC connection and logon to B1 without having to re-enter user and password.

SM59 to create a trusted RFC

Ltr to configure the connection

40. How you can use Data provisioning in SAP HANA studio?

Go to SAP HANA Modeler Perspective → Data Provisioning.

41. While creating data store in Data Services for SAP ECC, what is the datastore type in drop down list?

You have to select SAP Applications in data store type and SAP HANA in Database drop down list.

42. How you can perform trace, monitoring, error and performance monitoring of jobs in Data Services?

Replication job can be performed in Data Services Management Console. You have to go to status tab and select the repository where job is created → Batch Job Status.

There you can find different tabs- Trace, Monitor, Error and Performance Monitor.

43. How can you enable auditing of batch jobs in Data Services?

To enable auditing of batch job, this can be done in job execution parameters.

Go to status tab and select the repository where job is created → Batch Job configuration → Add Schedule.

Here you can find job execution parameters while adding a schedule for the job.

44. While configuring a batch job in SAP Data Services repository, what does owner name means?

Owner represents schema name where tables will be moved using batch job.

45. What are the different authentication methods configured in SAP HANA system for user creation?

Below is the list of authentication methods supported by SAP HANA –

- User name/Password
- Kerberos
- SAML 2.0
- SAP Logon tickets
- X.509

46. What is a restricted user in SAP HANA system?

Restricted users are those users who access HANA system with some applications and they don't have SQL privileges on HANA system. When these users are created they don't have any access initially.

If we compare restricted users with Standard users –

- Restricted users can't create objects in HANA database or their own Schemas.
- They don't have access to view any data in database as they don't have generic Public role added to profile like standard users.
- They can connect to HANA database only using HTTP/HTTPS.

47. What system privilege is required for a user to enable him to create user and roles in HANA studio?

Only database users with the system privilege ROLE ADMIN are allowed to create users and roles in HANA studio.

48. What are the different privilege types that can be assigned to a user in HANA studio?

- System Privilege
- Object Privilege
- Analytic Privilege
- Package Privilege
- Application Privilege
- Privilege on roles

49. Which of the system privilege is required to perform auditing related work?

Audit Admin

50. How you can configure Auditing and password policy in HANA studio?

Go to Security option in HANA system → Auditing

Under Global Settings→set Auditing status as enabled. You can also choose different Audit trail targets.

51. How you can connect SAP HANA to Microsoft Excel?

MDX Provider is used to connect MS Excel to SAP HANA database system. It provides driver to connect HANA system to Excel and is further used for data modelling. You can use Microsoft Office Excel 2010/2013 for connectivity with HANA for both 32 bit and 64 bit Windows.

52. How you can limit the size of data backup files in HANA system?

This can be done in File Based data backup settings. In Backup tab, go to Configuration → Limit Maximum file Size check box and enter the file size.

53. What are different recovery mechanism can be performed in HANA system?

Most Recent State – Used for recovering the database to the time as close as possible to the current time. For this recovery the data backup and log backup have to be available since last data backup and log area are required to perform the above type recovery.

Point in Time – Used for recovering the database to the specific point in time. For this recovery the data backup and log backup have to be available since last data backup and log area are required to perform the above type recovery.

Specific Data Backup – Used for recovering the database to a specified data backup. Specific data backup is required for the above type of recovery option.

Specific Log Position – This recovery type is an advanced option that can be used in exceptional cases where a previous recovery failed.

Note – To run recovery wizard you should have administrator privileges on HANA system.

54. Which all Business Objects tools can be directly connected to HANA Information views?

- SAP Lumira
- Analysis edition for OLAP
- SAP Crystal Reports
- Design Studio

55. What are the different modes in HANA studio that can be used for migration?

- Change and Transport system
- Delivery Unit
- Developer Mode
- SAP support mode

56. Which of the export options allows you to export multiple packages locally or to other HANA system?

Delivery Unit

Delivery unit is a single Unit which can be mapped to multiple packages and can be exported as single entity so that all the packages assigned to Delivery Unit can be treated as single unit.

57. How does SAP HANA support import from a local file?

Go to File → Import, You will see below option

This is used to import data from a flat file like .xls or.csv file. Click on Next → Choose Target System → Define Import Properties.

Select Source file by browsing local system. It also gives an option if you want to keep the header row. It also gives an option to create a new table under existing Schema or if you want to import data from a file to an existing table

SAP HANA Interview Questions and Answers

1. What are some tasks performed by Modeling Studio?

- Check database tables and retrieve META DATA
- Monitoring data transfer from SAPM Business Warehouse to other systems
- Schedule data replication jobs
- Utilize data services for modelling

2. **What is Latency?**

The total stretch of time consumed to replicate data from source system to target system is calculated as Latency.

3. **What is Row based and Column based approach?**

The database table is populated in the sequence of rows. Relational databases till now have been following Row based approach. On the contrary, Column based approach has tables being populated in sequence of columns occupying all contiguous memory locations. HANA supports both but is optimized more for Column approach.

4. **What is SLT? How do you configure it?**

It stands for SAP Landscape Transformation. It triggers replication of data from source to target system where either of the two ends has to be SAP HANA.

The configuration is performed in the Configuration & Monitoring dashboard using t-code LTR.

5. **What are the prerequisites to configure SLT?**

For SAP source systems – The role IUUC_REPL_REMOTE is assigned to the RFC user DMIS add-on has to be installed in the SLT system. For non-SAP systems – DMIS is not essential here. A database user can be authorized to perform replication activities.

6. **What are the different Replication techniques?**

- Trigger based replication using SLT
- ETL based replication using BODS
- Extractor data acquisition using DXC

7. **What are Transformation rules?**

These are the rules defined in the Advanced Replication Settings such that the data for the source tables is transformed during replication process. Fill Empty Fields

Convert Fields

Skip Records

8. **What factors affect the change in number of jobs?**

Number of Jobs depends upon the following –

- The number of tables to be replicated
- Number of configurations managed in SLT system
- Initial Load speed
- Expected Replication Latency time

9. **What are the different scenarios in Replication?**

Load – Starts the initial load of replication of data from the source system. It is a one-time process.

Replicate – Actual phase of replication. Database triggers and different logging tables for every table is created in SLT.

Stop – Stops any replication currently in process. Database triggers are aborted.

Suspend – Table replication is paused. Database trigger will not be deleted and changes will be filled in logging tables.

Resume – Restarts the replication for a suspended table.

10. **What is the maximum table name length and the Max number of columns allowed?**

Max table name length – 127 characters

Max no of columns – 1000 All geared up for a career in SAP HANA? Stay tuned and keep updated with more insights.

1. **What is SAP HANA?**

SAP HANA (High-Performance Analytic Appliance).

SAP HANA is SAP AG's implementation of in-memory database technology.

There are four components within the software group:

SAP HANA DB (or HANA DB) refers to the database technology itself,
SAP HANA Studio refers to the suite of tools provided by SAP for modeling,
SAP HANA Appliance refers to HANA DB as delivered on partner certified hardware (see below) as an appliance. It also includes the modeling tools from HANA Studio as well replication and data transformation tools to move data into HANA DB,
SAP HANA Application Cloud refers to the cloud based infrastructure for delivery of applications (typically existing SAP applications rewritten to run on HANA).

2. What is SAP HANA Appliance 1.0?

SAP HANA 1.0 is an analytics appliance that consists of certified hardware, an In Memory DataBase (IMDB) an Analytics Engine and some tooling for getting data in and out of HANA. You build the logic and structures yourself, and use a tool e.g. SAP BusinessObjects, to visualise or analyse data.

3. What are the product names?

The short answer is: it's a mystery. SAP has changed them around a lot and now they call it SAP HANA Appliance, SAP HANA Database and SAP HANA Studio. Applications built on HANA will be marked "powered by SAP HANA". Probably they will change it all again.

4. What additional limitations does Sybase Replication Server present?

SRS has additional restrictions which are worth bearing on mind. It can only replicate Unicode data and does not support IBM DB2 compressed tables.

5. What source databases does HANA support for batch loads?

If you use SAP BusinessObjects Data Services 4.0 for bulk loads then pretty much anything. BO-DS is a very flexible Extract, Transform & Load tool that supports many databases – check out the specs for more details.

6. What source databases does HANA support in real-time?

If you use Sybase Replication Server (SRS) for near real-time data then you need to watch out for licensing still (SAP have license deals pending). If you run DB2 then you're fine but with Oracle and Microsoft SQL Server there are some license challenges if you buy your license through SAP, because you may have a limited license that does not allow extraction. Talk to SAP for further information on this.

7. What storage subsystem does HANA use?

This varies from vendor to vendor but it is shared network attached storage (NAS). Both regular magnetic disks and SSD storage can be used for the backup of the database (HANA runs in memory remember, so disk storage is just for backup, and later, for data ageing). Note that you require 2x storage that you have RAM, which is 2x the database size – i.e. storage size = 4x database size. In most cases there is additional ultra-high speed SSD storage for log files.

8. How big does HANA scale?

Theoretically at least – very well. The biggest single-server HANA hardware will run most mid-size workloads – 2TB of in-memory storage is equivalent to 5-20TB of Oracle storage. The way that HANA works means that it is possible to chain multiple systems together – meaning that scalability has thus-far been determined by the size of customers' wallets. Do note that whilst SAP talk up "Big Data" quite a lot, HANA currently only scales to the small-end of Big Data, which refers to the kind of huge datasets that FaceBook or Google have to store – not Terabytes, but rather Petabytes.

9. Does SAP make their own IMDB/HANA hardware?

Yes, but only in the labs so far. There are no public plans to compete against IBM/HP/Dell in this space, but it may make sense for SAP to enter the appliance market, especially in the context of Data Centres and even more so in the context of the SAP Business byDesign cloud offering, which will run on IMDB.

10. Why doesn't HANA run on blades?

It's unclear but probably because the blades don't yet offer the same performance. HANA is optimized for the Intel X7560 CPU and will run fastest on this. And for instance, the Dell M910 blade can only run 2x X7650 CPUs and 512Gb RAM in this configuration, which probably explains the limitations. What's certain is that HANA will eventually run on blades – it's born to run on blade technology!

11. What hardware is supported right now?

Talk to your hardware vendor – all of the major vendors e.g. HP, IBM, Dell, have HANA offerings now. Technically HANA will run on any Intel x64 based system from your laptop through to the big 40-core, 2TB RAM servers. It is however only supported on a small number of big rack-mount servers like the Dell R910 and HP DL980.

12. What's the wider market opportunity for IMDB?

This is the interesting thing – no one knows yet, and few analysts seem to have cottoned on that the wider market opportunity might be huge. Think not just SAP applications but any third party that requires ultra-high speed. Think not just an appliance but a development platform. Time will tell.

13. You mean I have to buy a HANA only 2.5x smaller than my big Oracle RDBMS? What about archiving and data ageing?

Yes, in some instances you may have to buy a HANA appliance that is only 2.5x smaller than it would be under Oracle. And data ageing isn't part of the 1.0 release, but SAP is certainly working on it pretty hard. Let's hope they release something faster than you need to buy a bigger HANA appliance!

14. Does HANA/IMDB replace Oracle?

It's the elephant in the room, but once the Business Suite runs on IMDB, Oracle won't be needed any more by SAP customers who purchase HANA. This doesn't affect anything in the short term because those people buying HANA today will still need an Oracle ERP system.

15. Why is HANA so fast?

Regular RDBMS technologies put the information on spinning plates of iron (hard disks) from which the information is retrieved. HANA stores information in electronic memory, which is some 50x faster (depending on how you calculate). HANA stores a copy on magnetic disk, in case of power failure or the like. In addition, most SAP systems have the database on one system and a calculation engine on another, and they pass information between them. With HANA, this all happens within the same machine.

16. What does HANA cost?

SAP hasn't entirely confirmed HANA licensing costs but the hardware is somewhere around \$1-200k per TB. Add to this licensing costs which are still being made on a per-customer basis.

17. What is HANA bad at?

There are some current issues around HANA when delivering ad-hoc analytics, especially when using the SAP BusinessObjects Webi tool. Essentially the problem is that you can ask computationally very difficult questions with Webi, which can cause very long response times with HANA. SAP will need to build optimization for both Webi and HANA to reduce the computational complexity of these questions, but they're not there yet.

What's more, it's worth noting that HANA 1.0 is not a Data Warehouse and it is more of a Data Mart – that is, suited to point applications where there is a clear use case.

18. What is HANA great at?

The best thing that HANA brings to the table is the ability to aggregate large data volumes in near real-time – and to have the data updated in near real-time. SAP's demos show hundreds of billions of records of data being aggregated in a matter of seconds. SAP has built a set of Analytics Apps on top of HANA and these are set to be great point use cases to get customers up and running quickly.

19. If I can run NetWeaver BW on IMDB/HANA, why can't I run the Business Suite/ERP 6.0?

Simply because it's not mature enough yet to support business critical applications. From a technology perspective, it is already possible to run the Business Suite on IMDB and SAP has trialled moving some large databases into IMDB already.

20. What's the difference between HANA and IMDB?

HANA is the name for the current BI appliance (HANA 1.0) and the BW Data Warehouse appliance (HANA 1.0 SP03). Both of these use the SAP IMDB Database Technology (SAP HANA Database) as their underlying RDBMS. Expect SAP to start to differentiate this more clearly as they start to position the technology for use cases other than Analytics.

21. What is SAP HANA 1.5, 1.2 or 1.0 SP03?

These are all the same thing, and 1.0 SP03 is touted to be the final name for what should go into RampUp (beta) in Q4 2011. This will allow any SAP NetWeaver BW 7.3 Data Warehouse to be migrated into a HANA appliance. HANA 1.0 SP03 specifically also accelerates BW calculations and planning, which means you get even more performance gains.

22. What are the limitations of HANA 1.0?

Quite a few so far – it can only replicate certain data, from certain databases, in certain formats, using the Sybase Replication Server. Batch loading is done using SAP BusinessObjects Data Services 4.0 and is optimised only for SAP BusinessObjects BI 4.0 reporting.

23. What are the product names?

The short answer is: it's a mystery. SAP has changed them around a lot and now they call it SAP HANA Appliance, SAP HANA Database and SAP HANA Studio. Applications built on HANA will be marked "powered by SAP HANA". Probably they will change it all again.

24. Define Five-minute rule?

It is a rule of thumb for deciding whether a data item should be kept in memory, or stored on disk and read back into memory when required. The rule is "randomly accessed disk pages of cache are re-used every 5 minutes".

25. Define multi-core CPU?

Multiple CPU's on one chip or in one package is called multi-core CPU.

26. Define Stall?

Waiting for data to be loaded from main memory into the Cpu cache is called as Stall.

27. What is SAP In-Memory Appliance (SAP HANA)?

HANA is an in-memory technique to store data that is particularly suited for handling very large amounts of tabular, or relational, data with extra ordinary performance. Common databases store tabular data row wise. Reorganizing the data in memory column-wise brings a tremendous speed increase when accessing a subset of the data in each table row.

28. What are the components or products of HANA?

SAP HANA contains the following components.

- SAP HANA DATABASE
- SAP HANA Studio SAP HANA CLIENT
- SAP HANA INFORMATION COMPOSER

– DIAGNOSTICAGENT7.3

- SAP HANA client package for MS excel
- SAP HANA UI for Information Access (INA)
- SAP HANA A F L L O

Software Update Manager for SAP HANA

- SAP LT Replication Add On
- SAP LT Replication Server
- SAP HANA Direct Extractor Connection (DXC)
- SAP Data Services 4.0

29. What are the different editions available in HANA appliance software?

Platform Edition:

Platform edition is intended for customers who want to use ETL-based replication and already have a license for SAP BO Data Services.

Enterprise Edition:

Enterprise edition is intended for customers who want to use either trigger-based replication or ETL-based replication and do not already have all of the necessary licenses for SAP BO Data Services.

30. . What is columnar and Row-Based Data Storage?

A database table contains data in the form of rows and columns. However Computer memory is organized as a

linear structure. To store a table in linear memory, there are two options. A row-based storage stores a table as a sequence of records, each of which contains the fields of

one row. In a columnar storage the entries of a column are stored in contiguous memory locations. The SAP HANA database allows specifying whether a table is to be stored column-wise or row-wise. It is also possible to alter an existing table from columnar to row-based and vice versa. Search operations in tabular data can be accelerated by organizing data in columns instead in rows.

31. What are the advantages of Column based tables?

Calculations are typically executed on single or a few columns only.

The table is searched based on values of a few columns.

The table has a large number of columns.

The table has a large number of rows and columnar operations are required (aggregate, scan, etc.).

High compression rates can be achieved because the majority of the columns contain only few distinct values (compared to number of rows).

32. What are the advantages of Row-based tables?

The application needs to only process a single record at one time (many selects and/or updates of single records).

The application typically needs to access a complete record (or row).

The columns contain mainly distinct values so that the compression rate would be low.

Neither aggregations nor fast searching are required.

The table has a small number of rows (e. g. configuration tables).

33. Which case the data to be stored in columnar storage?

To enable fast on-the-fly aggregations, ad-hoc reporting, and to benefit from compression mechanisms it is recommended that transaction data to be stored in a column-based table.

34. What is paralisation?

Column-based storage makes it easy to execute operations in parallel using multiple processor cores.

In a column store

data is already vertically partitioned means that operations on different columns can easily be processed in parallel. If

multiple columns need to be searched or aggregated, each of these operations can be assigned to a different processor core. In addition operations on one column can be parallelized by partitioning the column into multiple sections that can be processed by different processor cores.

35. What are the different Compression Techniques?

1. Run-length encoding

2. Cluster encoding

3. Dictionary encoding

36. Why materialized aggregates are not required?

With a scanning speed of several gigabytes per millisecond, in-memory column stores, make it possible to calculate aggregates on large amounts of data on the fly with high performance. This is expected to eliminate the need for materialized aggregates in many cases.

37. What are the advantages of Eliminating materialized aggregates?

Simplified data model Simplified application logic Higher level of concurrency and With the fly Aggregation
we have aggregated values up to date

38. What are the different types of replication techniques?

- ETL based replication using BODS
- Trigger based replication using SLT
- Extractor based data acquisition using DXC.

39. Define SLT?

SLT stands for SAP [andscape Transformation which is a trigger based replication. SLT replication server is the replication technology to pass data from source system to the target system. The source can be either SAP or non-SAP. Target system is SAP HANA system which contains HANA database.

40. What is Configuration in SLT?

The information to create the connection between the source system, SLT system, and the SAP HANA system is specified within the SLT system as a Configuration. You can define a new configuration in Configuration & Monitoring Dashboard (transaction LTR).

41. What is Configuration and Monitoring Dashboard?

It is an application that runs on SLT replication server to specify configuration information (such as source system, target system, and relevant connections) so that data can be replicated, it can also use it to monitor the replication status (transaction LTR).

Status Yellow: it may occur due to triggers which are not yet created successfully.

Status Red: it may occur if master job is aborted (manually in transaction SM37).

42. What is advanced replication settings?

A transaction that runs on SLT replication server to specify advanced replication settings like

- a. Modifying target table structures.
- b. Specifying performance optimization settings
- c. Define transformation rules

43. Define Latency?

It is the length of time to replicate data (a table entry) from the source system to the target system.

44. Define logging table?

A table in the source system that records any changes to a table that is being replicated. This ensures that SLT replication server can replicate these changes to the target system.

45. What are Transformation rules?

A rule specified in the Advanced Replication settings transaction for source tables such that data is transformed during the replication process.

Example you can specify rule to:

- i. Convert fields
- ii. Fill empty fields
- iii. Skip records

46. When to change the number of Data Transfer job?

If the speed of the initial load/replication latency time is not satisfactory 1f SLT replication server has more

resources than initially available, we can increase the number of data transfer and/or initial load jobs. After the completion of the initial load, we may want to reduce the number of initial load jobs.

47. When to go for table partitioning?

1f the table size in HANA database exceeds 2 billion records, split the table by using portioning features by using "Advanced replication settings" (transaction IUUC_REPLCONT, tab page IUUC_REPLTABSTG).

48. What are the jobs involved in replication process?

- Master Job (IUUC_MONITOR_)
- Master Controlling Job (IUUC_REPLIC_CNTR_)
- Data Load Job (DTL_MT_DATA_LOAD_< MT_ID >_<2digits>)
- Migration Object Definition Job(IUUC_DEF_MIG_OBJ_<2digits>)
- Access Plan Calculation Job(ACC_PLAN_CALC__<2digits>)